Idle gear. It is also useful in changing the direction of rotation of the follower with out changing the speed.

In simple gear train, if the total number of gears are in odd number then the first and last gear rotates in the same direction

Example 6

Gears A,B,C and D are connected by a simple gear train. The number of teeth on them are 75, 45, 60 and 50. If the gear D rotates at a speed of 360 rpm in clock-wise direction, what will be the speed of the gear A.

T_1 = 75 teeth T_2 = 45teeth
T_3 = 60 teeth T_4 = 50teeth

Number of teeth on the driving gear (T_1) = 75 teeth

Number of teeth on the ideal gear (T_2) = 45 teeth

Number of teeth on the ideal gear (T_3) = 60 teeth

Number of teeth on the driver gear (T_4) = 50 teeth

Number of rotational of driven gear N_4 = 360 rpm

$$T_1 N_1 = T_4 N_4$$

$$N_1 = \frac{T_4 \times N_4}{N_1}$$

$$N_1 = \frac{50 \times 360}{75}$$

$$N_1 = \frac{2 \times 360}{3} = 240 \text{ rpm}$$

N_1 = 240 RPM Anti clock wise direction

COMPOUND GEAR TRAIN

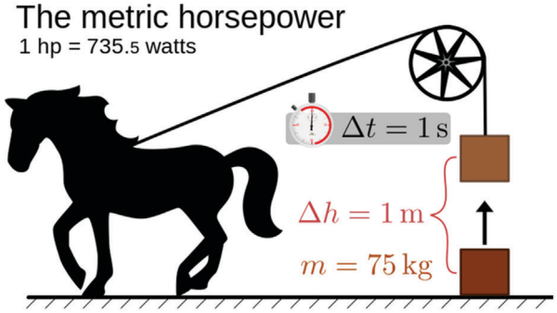


If the gear drive is arranged by keeping more than one gear on a shaft , it is called compound gear train. The net velocity ratio of the gear drive is influenced by the intermediate gear also, so it is possible with a compound gear train to attain a higher velocity ratio in limited space. The direction of rotation of the follower with respect to the driver is determined by a number of intermediate gears on separate shafts.



Mechanical Horse Power
1 HP= 745.7 Watts

The metric horsepower
1 hp = 735.5 watts



Horse power (hp) is a unit of measurent of power (the rate at which work is done). Two common definitions being used to day are the mechanical horse power or emperial horse power, which is 745.7 watts, and the metric horse power, whcih is apprximately 735.5 watts.

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